Fenestra Site

User Manual

Fenestration installation planning, delegation, quality, commercial and payment control, from BOQ to monthly IPA, with monthly KPIs and recognition.

Edition for version 23. Covers the Installation module, the Engineering module and the common tabs.

This manual follows the working order of a real project. Chapters 1 to 10 are the day-to-day workflow. Chapters 11 to 16 describe the newer controls: the document register, off-times and forecast, delegation alarms, rejected material reports, scan locations, and KPIs with certificates. Chapters 17 to 19 describe the Engineering module: roles and working order, screens and daily use, and engineering KPIs. The appendices cover the common tabs, other tabs, alarms, roles and administration.

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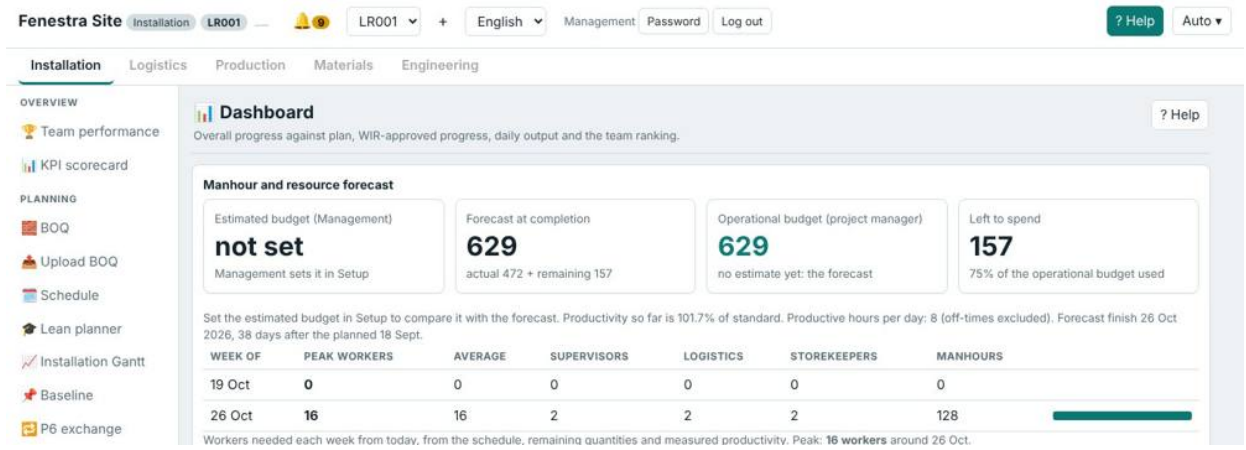
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How the application works

Fenestra Site holds one project at a time (the project selector is at the top). Everything you do is saved in the database, logged with your name and the time, and shared with the other people on the project. You only see the tabs your role is allowed to use.

The three navigation areas

Area	What it holds
Module tabs (header, under the title)	Installation, Logistics, Production, Materials and Engineering. Chapters 1 to 16 describe the **Installation module**; chapters 17 to 19 describe the **Engineering module**. The Logistics, Production and Materials tabs are greyed out until those modules are built.
Left menu (Installation tabs)	Grouped by job: Overview (Team performance, KPI scorecard), Planning, Resources, Site execution and Commercial. A tab appears only if your role may open it.
Footer tabs (bottom of every page)	The common tabs that are not specific to one module: Dashboard, Tasks, Inbox, Calendar, Chat, Alerts, Documents, Library, Reports, Contacts, Portfolio, Activity log, Audit, Users, Help and Setup. See appendix A.



The header with module tabs, the left menu and the footer tabs (Dashboard showing the manhour and resource forecast).

The whole project in ten steps

#	Step	Main tab(s)	Who
1	Upload the BOQ and the high-level installation P6 programme	Upload BOQ, P6 exchange	Project manager, planning engineer
2	Create sub-BOQ items (building, level, elevation)	Upload BOQ, P6 exchange	Project manager
3	Modify the schedule including the sub-BOQ items	Schedule, Lean planner, Installation Gantt	Project manager, planning engineer
4	Delegate tasks: alerts from the schedule, pre-start checklist, send to site	Look-ahead, Delegate resources, Task delegation	Project manager
5	Site engineer reviews and accepts or rejects tasks	Site (Invitations)	Site engineer
6	Delegate to team leaders, clock in and out, daily results	Site, QR badges	Site engineer, team leaders
7	Materials received, QA/QC check, MIR raised	Materials	Site engineer, quality engineer
8	Task completion, QA/QC check, WIR raised, grouped and sent	Site, WIR log	Site engineer, quality engineer
9	Record obstructions, instructions, snags and requests	Site, Instructions, Snagging, Requests	Everyone on site
10	Weekly reports, monthly IPA and delay claims	Reports, Rates and IPA, Delay analysis	Project manager, Management

Signing in

1	Open the web address of your installation and enter your user name and password.
2	The first time, you must choose a new password of at least 8 characters.
3	New staff can press Sign up on the login screen. Their request waits in Setup for Management to approve and to confirm the role.
4	Use Password (top right) to change your password and Log out when you leave a shared device.

Getting help

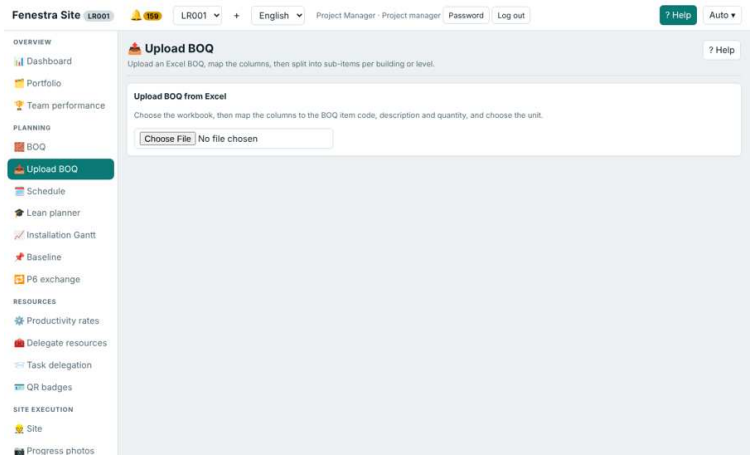
Press **? Help** at the top right on any page for the guide for that page, a list of all topics and common questions. Press Ctrl+K to search. The **Help** footer tab holds the full guides, a quick-start checklist for your role and a printable version. The language selector offers English, Arabic, Hindi, Urdu, Filipino, Bengali, Tamil and Chinese; the layout switch at the top right changes between PC, iPad and phone views.

1. Upload the BOQ and the high-level P6 programme

Who: project manager, planning engineer (Management can do it too). Where: Upload BOQ, then P6 exchange.

Upload the BOQ

1	Open Upload BOQ and choose the Excel file of the bill of quantities.
2	Map the columns: item code, description and quantity, and choose the unit (m2, lm or nr).
3	Check the preview. A wrong column mapping is the most common mistake.
4	Confirm to load the BOQ. It appears in the BOQ tab as a tree with the live % complete and the WIR-approved % for each item.



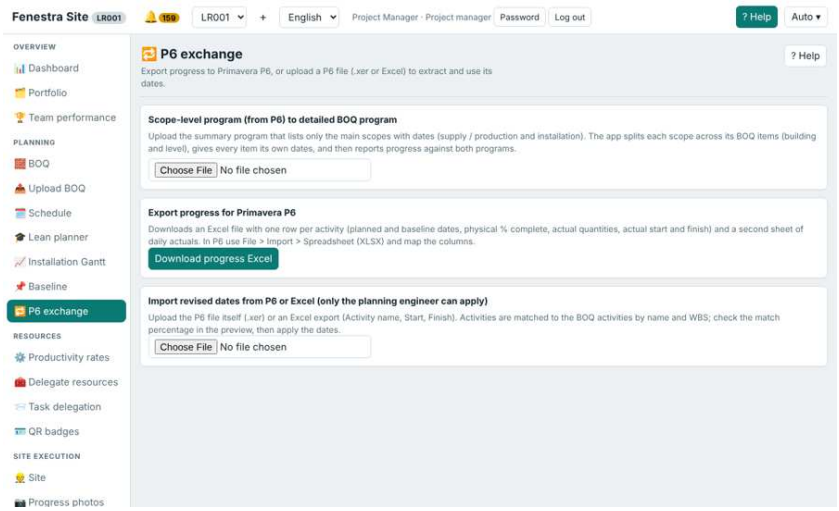
Upload BOQ: choose the file, map the columns and confirm.

Upload the high-level installation programme from P6

If your planner has a summary Primavera P6 programme that lists only the main scopes (supply or production, and installation, each with a start and finish), you do not need to type dates.

1	Open P6 exchange. Under Scope-level program choose the .xer or Excel file.
2	Pick the building (the WBS node) in the file and whether to use the current (early) dates or the planned (target) dates.
3	The app matches the activity names to the BOQ scopes and tells you how many matched. Names that do not match are listed.
4	Press Generate detailed program. Every BOQ item gets its own dates inside the window of its scope. The table shows the crew effort against the standard productivity: over 100% means the window is shorter than the standard allows, so more crews are needed.
5	Press Apply detailed program to the schedule to re-date the installation activities. A note is written in the activity log.

Progress is then reported at both levels: against the main-scope program and against each BOQ item, with planned %, actual %, variance and a forecast finish. You can download the result as Excel, and a detailed .xer for P6 (built by the server). You can also use the ordinary P6 import (same tab) to take dates from a P6 file and apply them to matching activities after a preview.



P6 exchange: upload the scope-level programme and generate the detailed program.

2. Create sub-BOQ items

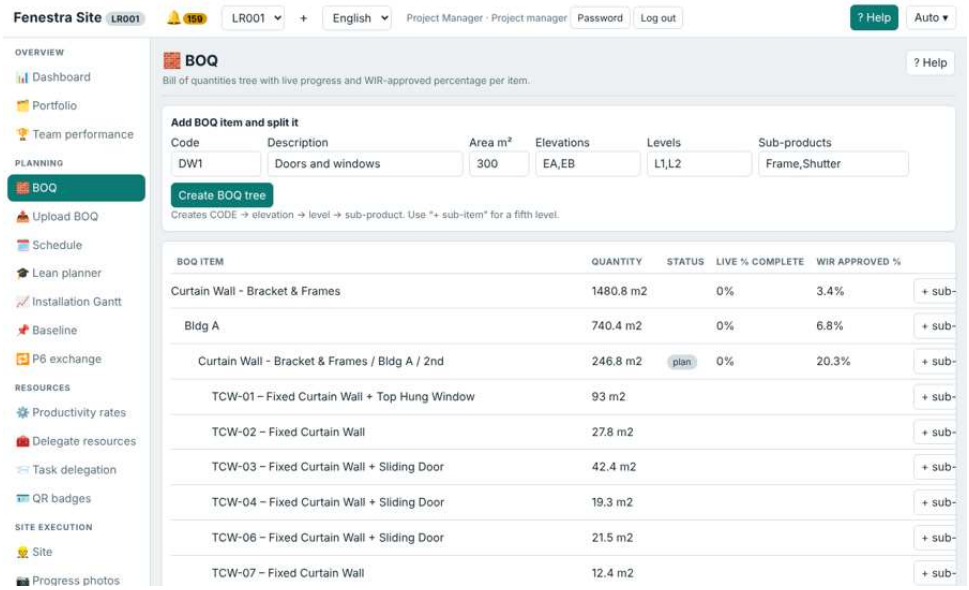
Who: project manager. Where: Upload BOQ (pop-up) or the BOQ tab.

A BOQ line such as Curtain wall frame must be split into buildings, levels or elevations so that work can be planned, delegated and inspected where it actually happens.

1	After importing the BOQ, the Add sub-BOQ items pop-up opens. Tick the BOQ lines to split.
2	Enter the elevations (for example EA, EB), the levels (for example 2nd, 3rd, 4th) and the sub-products, separated by commas. Leave a box empty to skip that dimension. Press Skip if a line should not be split.
3	The quantity of the line is shared equally between the resulting sub-items (elevation x level x sub-product). You can edit the quantities afterwards.
4	Each sub-item becomes its own activity in the schedule, with the standard duration worked out from the productivity rate, and the schedule is recalculated.

In the BOQ tab you can also use Add BOQ item and split it (code, description, area, elevations, levels, sub-products) to create a line with its sub-items in one step, and the + sub button beside any line to add a sub-item by hand.

Tip: keep the same level and elevation names everywhere. The Lean planner, progress photos, snagging and reports group work by those names.

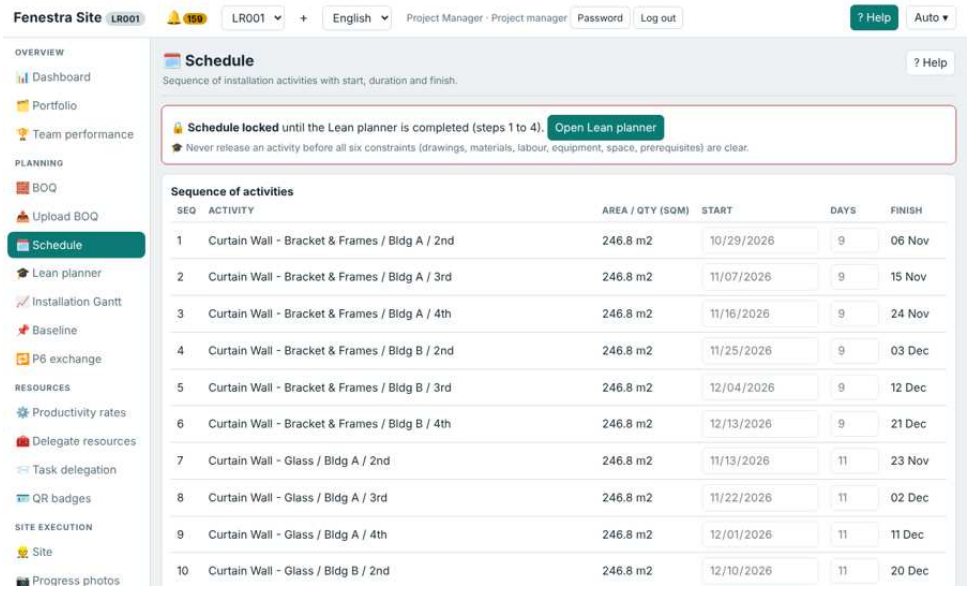


BOQ tab: live % complete and WIR-approved % for every sub-item.

3. Modify the schedule including sub-BOQ items

Who: project manager, planning engineer. Where: Schedule, Lean planner, Installation Gantt, Baseline.

1	Open Schedule to see every sub-BOQ item with start, duration, finish and the quantity.
2	Change a start date or duration, or move an activity earlier or later. If the Lean planner is switched on, the changes are made through the Lean planner so that each change gets a reason (see appendix B).
3	Check the result on the Installation Gantt.
4	When the programme is agreed, the planning engineer presses Freeze baseline in the Baseline tab. Later variances and extension of time claims are measured against it.



Schedule: start, duration and finish for every sub-BOQ item.

Rule of thumb: duration follows the standard productivity (quantity per team per day) in the Productivity rates tab, and the working day is the productive hours left after the off-times set in Setup (chapter 12). Management changes the rates, and changes may need approval.

4. Delegate tasks, alerted from the schedule, with the checklist

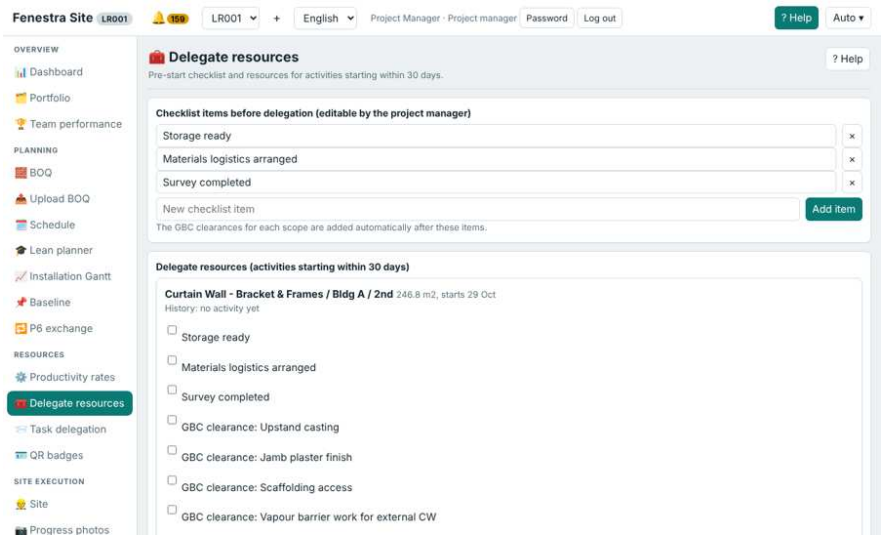
Who: project manager. Where: Look-ahead, Delegate resources, Task delegation.

1 **Look-ahead (30, 7 and 1 day).** 30 days before an activity starts, the project manager, site engineer and planner are notified (and emailed) with a one-month look-ahead plan, downloadable as Excel or as a printable PDF page. 7 days before, the project manager delegates each task to a site engineer with a checklist (resources, materials, access equipment such as cradle or crane, permits, work area) and the automatic weather forecast. 1 day before, the site engineer confirms with the same checklist that the next working day's work is doable, or says why not. The checklist items are editable by the project manager; set the site location and weather limits in Setup.

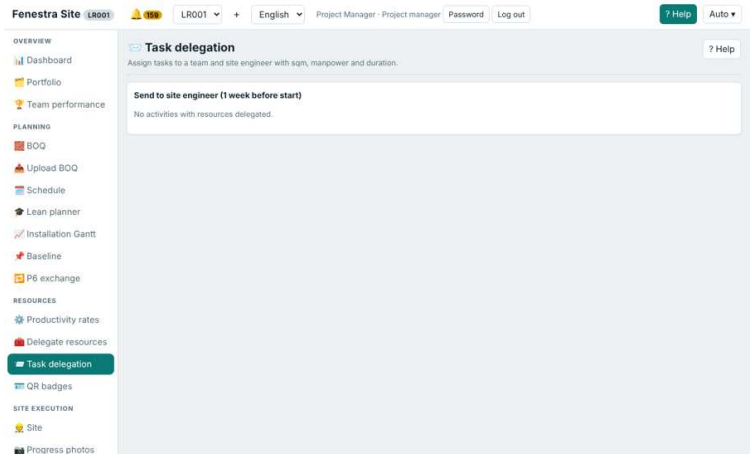
2 **Delegate resources.** For activities starting within 30 days, work through the pre-start checklist (editable by the project manager) and the resources and materials. Missing items automatically trigger an email to the Main contractor or the Production department.

3 **Task delegation.** One week before the start, choose the task and the team. The app shows the manpower and duration from the standard rate. Press send. The site engineer receives an invitation.

The checklist is what stops an activity being released before it is ready. If a prerequisite is missing the app tells you, and the missing item is recorded in the activity log with date and time. If the project manager does not delegate in time, the delegation alarms in chapter 13 escalate it.



Delegate resources: the pre-start checklist and resources.

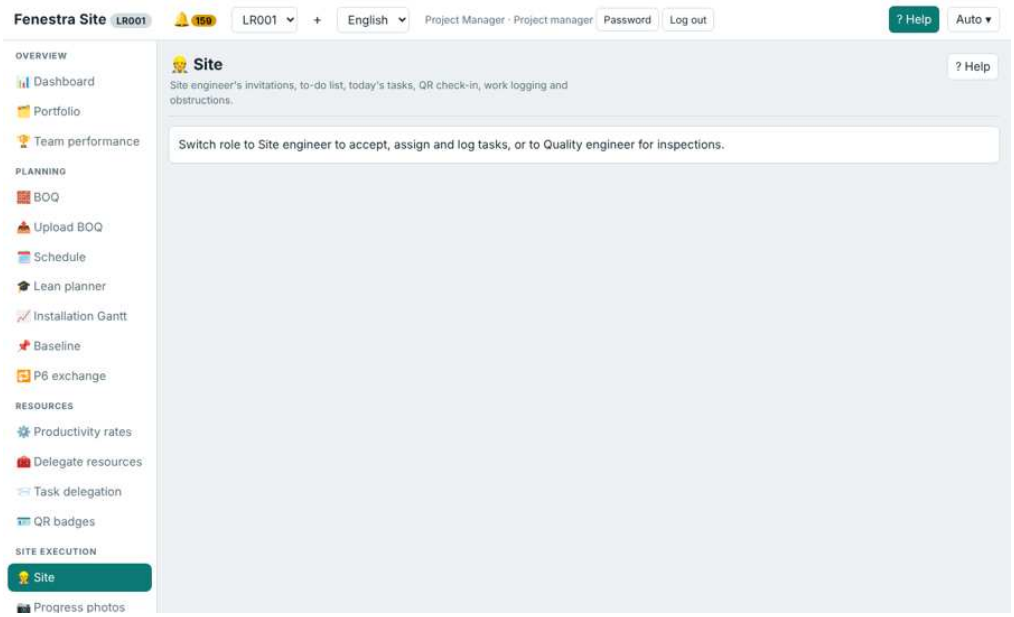


Task delegation: send ready activities to the site engineer.

5. Site engineer reviews and accepts or rejects tasks

Who: site engineer. Where: Site, Invitations.

1	Open Site. Each delegated task appears under Invitations with quantity, manpower, duration and the prerequisites.
2	Check on site that every prerequisite is really available.
3	Press Accept, or Revert with the list of missing items and a note. A revert goes back to the project manager and is logged.
4	Accepted tasks move into your to-do list and Today's tasks when their start date comes.



Site tab: invitations, to-do list and today's tasks.

6. Delegate to team leaders, clock in and out, daily results

Who: site engineer and team leaders. Where: Site, QR badges.

Set up once

1	In QR badges add the employees with their ID, role (team leader, installer, helper, foreman and so on) and team, then print or show each QR badge.

Every working day

1	Under Today's tasks choose the team for the task (each team is led by its team leader) and the number of workers.
2	Each worker scans his QR badge: this clocks him in. The scan also logs the device location (chapter 15). Press Start.
3	During the day record any obstruction (see chapter 9).
4	At the end of the day enter the quantity installed (m2) and the hours, then press Finish and log. This clocks the team out and records the day's result: quantity, workers, hours and manhours.

The system stops impossible entries: the cumulative quantity can never exceed the BOQ quantity.

Daily actual versus plan

The Dashboard shows, for every working day, the planned quantity, the actual quantity logged, the variance and the running totals, for the current week and the current month. Each day is closed automatically from 17:00 and its plan is kept, so later schedule changes do not alter past days.

Time sheets, warning letters and idle teams

Every QR scan on a task is the clock-in and the day's work log is the clock-out. Time sheets gives daily, weekly and monthly sheets with normal and overtime hours (shift start and end, normal hours per day, unpaid break, Sunday and holiday rules are set in Setup; the off-times of chapter 12 are deducted), downloadable as Excel or a printable PDF page. If someone assigned to a running task has not scanned in after the shift start plus the grace time, a warning letter (first, second, final) is issued automatically and logged in a downloadable warning log; the project manager can cancel a warning with a reason. When a team has no running task, the site engineer and project manager are notified and the Dashboard suggests suitable scheduled tasks.

Danger notices

The safety team raises a danger notice under Danger notices. It is emailed at once to every party with a contact in Setup, including the Main contractor, and every user sees a red alarm banner until they acknowledge it. The register is downloadable.

Contract terms, tools and logistics

In Setup, upload the Conditions of Contract (Word, PDF or text) and enter the clause numbers; they are used together with FIDIC in variation letters, EOT notices and delay claims. Thirty days before an activity starts, the look-ahead also asks for the tools requirement and reminds daily until it is ticked. Logistics teams and their productivity are set in Setup; the number of logistics teams needed is shown with each task, 'Logistics team assigned' is on the assignment checklist, and the project manager is reminded to add logistics sub-tasks when creating sub-items.

Rewards

Achievements are notified to the project manager and site engineer at once, but the reward is held until the WIR for the task is approved. The project manager then approves, and Management gives the final approval. The team-of-the-month award cannot be confirmed while its tasks are still waiting. The monthly KPI ranking and certificates of recognition are in chapter 16.

Alarm if no daily update is made

If a started activity is scheduled for the day but has no work logged, or the team was clocked in and not out, the app raises an alarm: for yesterday's missing updates at the first opening, and for today's from 17:00. The project manager and site engineers get a notification, and the project manager's dashboard shows a Daily updates missing list. Sundays are skipped.

Carried-over work and forecast finish

If a task passes its planned finish with quantity left, the app records Carried over, X m2 remaining for that day (once per task per day, Sundays skipped) and the same team keeps the task in Today's tasks, with the target capped at what is left. For every started or late task the Schedule and Installation Gantt show a forecast finish (remaining quantity divided by the team's actual daily output) and the delay against plan. The forecast changes nothing by itself. The project manager or planning engineer may tick tasks and press Apply forecast to schedule: durations are extended, unstarted successors that would finish before the extended task move later, the result is checked against the baseline completion, and an EOT letter to the Main contractor can be issued if it finishes later. If the Lean planner is on, a reason is required and is added to the lean change log.

7. Materials: QA/QC check, then MIR

Who: site engineer records, quality engineer approves, project manager assigns locations. Where: Materials.

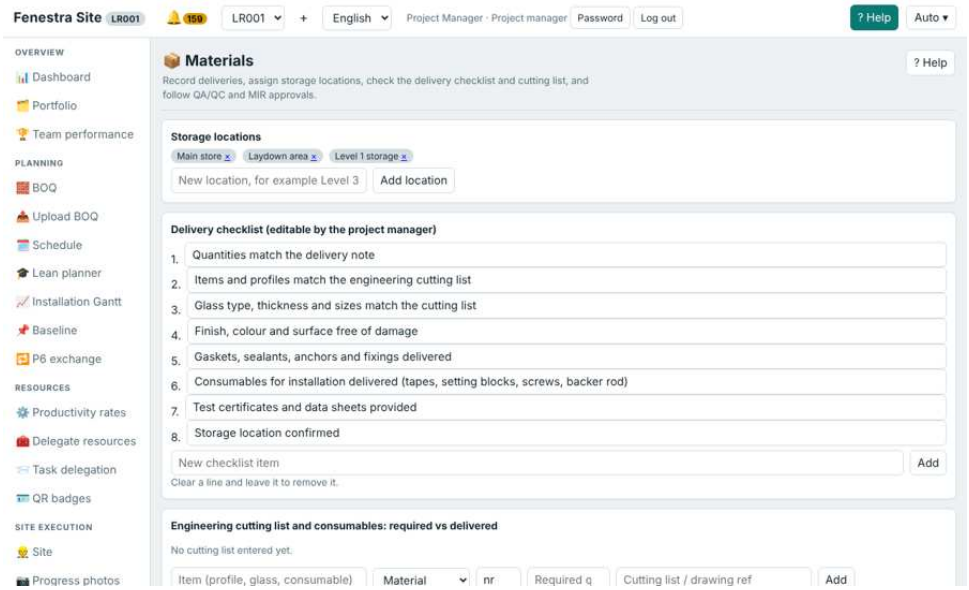
Set up once (project manager)

- Add the storage locations (main store, laydown area, levels).
- Edit the delivery checklist: quantities match the delivery note, items match the engineering cutting list, finish free of damage, consumables delivered, certificates provided, and so on.
- Enter or import the cutting list issued by engineering, including the consumables needed for installation.

For every delivery

1	The site engineer presses Record delivery: date, supplier, delivery note, material, the cutting-list item and BOQ line it belongs to, the quantity, and the equivalent BOQ quantity (used for the payment claim). Tick the checklist.
2	The project manager assigns the storage location.
3	The quality engineer inspects and presses QA approve or Reject. If checklist items are not ticked you are asked to confirm.
4	On approval an MIR (material inspection request) is raised automatically to the Main contractor with a formal letter that includes the checklist and the location. The MIR is serial-numbered and logged in the document register (chapter 11).
5	When the Main contractor replies, log the decision (Approved, Approved with comments, Rejected) with its date. A rejection raises a site observation report (chapter 14).

The table Engineering cutting list and consumables compares what was required with what was delivered and QA-approved, and shows Complete, Short or Not delivered for every item, so you can tell at a glance whether everything on the cutting list has arrived.



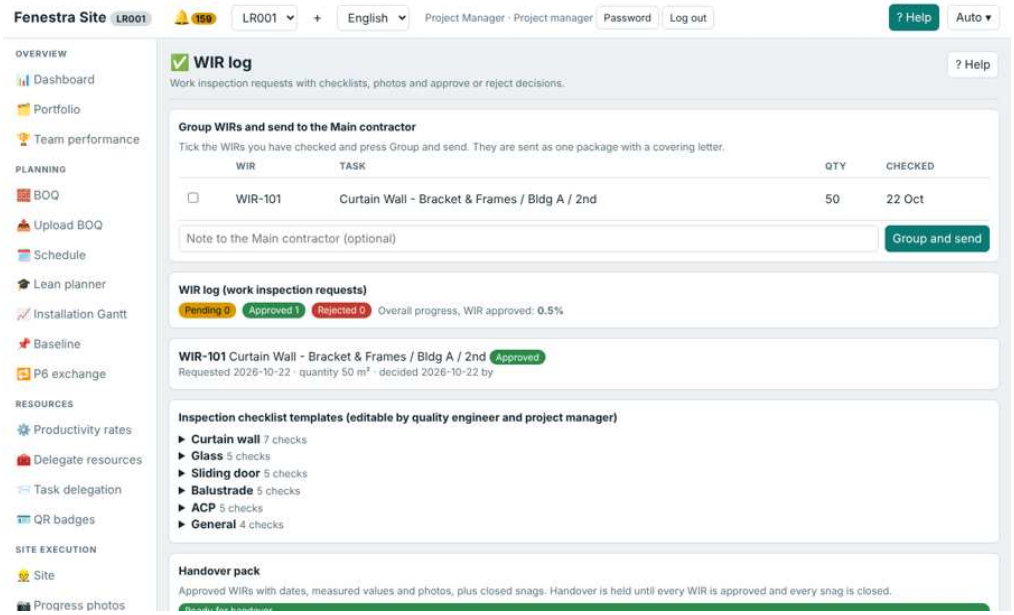
Materials: locations, checklist, cutting list, deliveries and MIR.

8. Task completion: QA/QC check, then WIR

Who: site engineer requests, quality engineer checks and groups. Where: Site, WIR log.

1	When a task or part of it is complete, the site engineer presses Request WIR for completed quantity (it is also raised automatically when a task reaches 100%).
2	In the WIR log the quality engineer opens the WIR, completes the inspection checklist (tick, pass/fail or value items) and adds photos.
3	Press Approve, or Reject with a reason and an optional photo. A rejected WIR is reworked and requested again.
4	Group and send: at the top of the WIR log, tick the WIRs that you have checked and press Group and send. They go to the Main contractor as one package (for example WIRP-001) with a covering letter listing every WIR, task and quantity.
5	Log the Main contractor's decision on the package (Approved or Rejected, with the date and comment).

In Setup you can choose whether a WIR counts in the IPA from your own QA/QC approval, or only after the Main contractor approves the package.



WIR log: inspections, approvals and grouping into packages.

9. Record obstructions, instructions and other site records

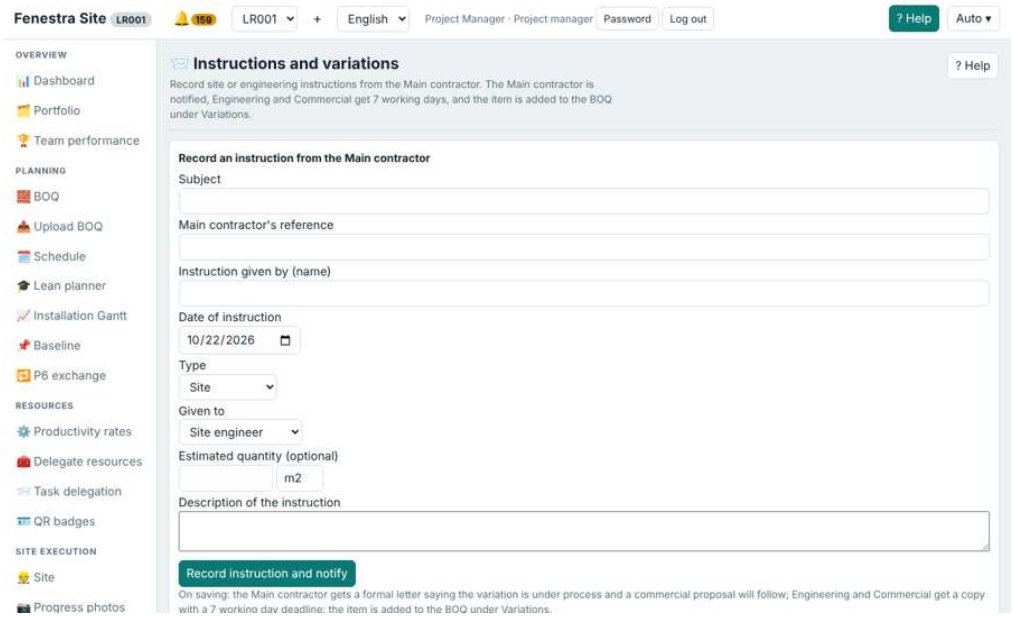
Obstructions (site engineer). In Site use Record obstruction: choose the task, the type, the hours lost, a note and photos. They feed the weekly delay analysis and the FIDIC delay report.

Instructions from the Main contractor. Open Instructions, enter the subject, who gave it, the date, whether it is a site or engineering instruction, and the description. Only Management and the project manager see and issue instructions, delay letters and EOT letters. When you record it:

- the Main contractor receives a formal letter saying the variation is under process and that a commercial proposal will follow;
- Engineering and Commercial receive a copy with a deadline of 7 working days;
- the item is added to the BOQ in a group called Variations.

The project manager then enters quantity, rate and time impact from Engineering and Commercial and issues the proposal letter, and records the Main contractor's decision. The project manager or planning engineer creates the schedule for the variation; if it finishes after the baseline completion date, an extension of time letter is issued automatically.

Other records. Requests to other parties (material, clearance) are in Requests with automatic escalation when overdue. Snagging holds the snag list with drawing pins, photos and close-out; the quality engineer can also record a snag or de-snag inspection approval there. Safety holds permits, toolbox talks and incidents. Progress photos collects weekly photos per building and elevation. Chat and the Activity log keep the project record.



Instructions: record, notify, price and schedule the variation.

10. Weekly reports, monthly IPA and delay claims

Who: project manager, Management. Where: Reports, Delay analysis, Rates and IPA, Commercial.

Reports

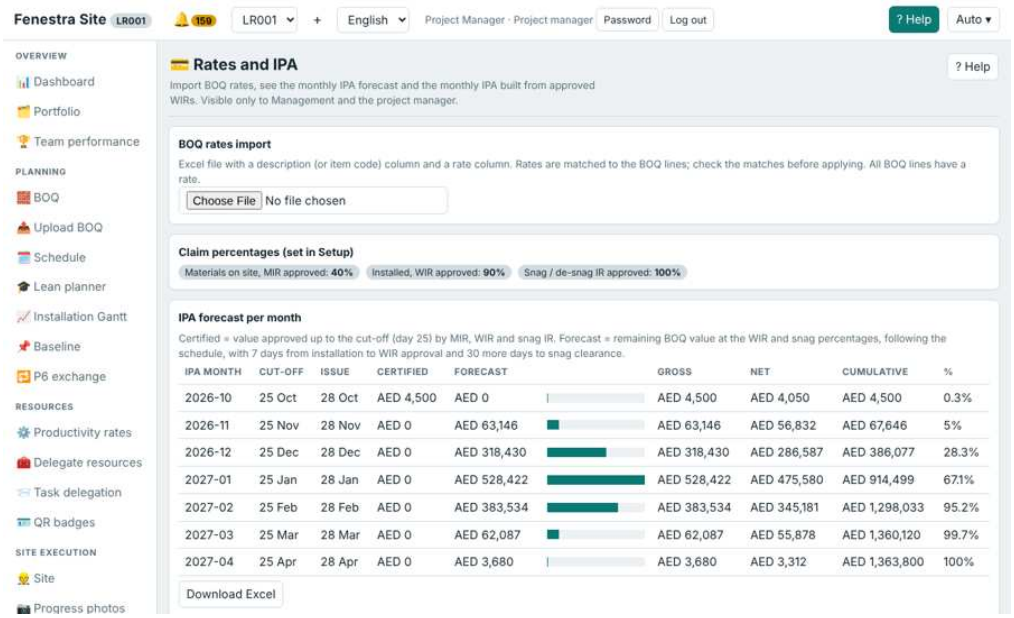
Reports holds the daily diary, weekly and monthly reports, which you can print or export to Excel. The Dashboard gives the live picture: plan versus actual, WIR-approved progress, the manhour and resource forecast (chapter 12), cumulative IPA against contract value, and WIR, MIR and variation orders raised versus approved.

Monthly IPA

1	In Setup the project manager sets the cut-off day and issue day of each month, and the claim percentages: materials on site with an approved MIR, installed work with an approved WIR, and work with its snag or de-snag inspection approved.
2	In Rates and IPA import the BOQ rates from Excel (visible only to Management and the project manager).
3	On the issue day a draft IPA is prepared automatically from the MIR, WIR and snag approvals up to the cut-off day, with retention. Review it and press Issue to Main contractor. The IPA is numbered and logged in the document register.
4	The same tab shows the IPA forecast per month, using the rates and the execution schedule, and the Excel download.

Delay analysis and claims

Delay analysis gives a weekly analysis with substantiation, and each month a delay report and extension of time claim letter based on FIDIC (1999 or 2017). The Main contractor's responses and the monthly notices are kept with the project record. Delay and EOT letters are numbered and logged in the document register.

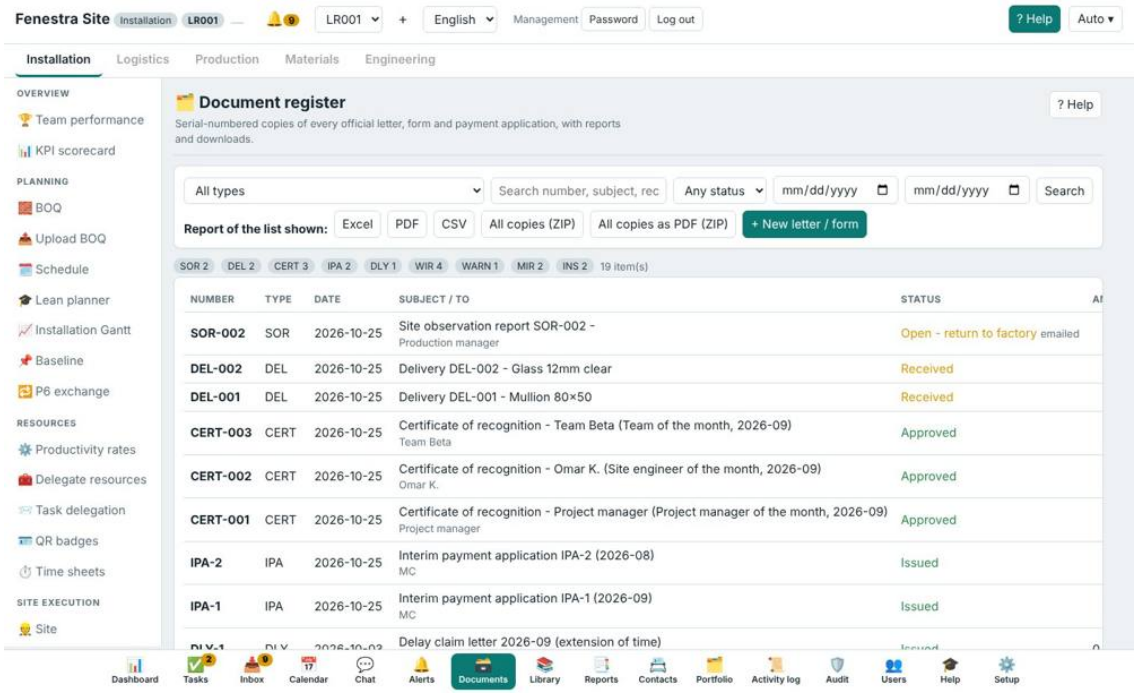


Rates and IPA: rates import, forecast and the monthly IPA.

11. Document register, numbering and formats

Who: everyone sees their own documents; Management and the project manager see commercial ones. Where: Documents (footer), Setup.

Every official letter and form is serial-numbered and logged with a stored copy: WIR, MIR, IPA, delay letters, EOT letters, variation proposals and orders, delivery notes, warning letters, site observation reports and certificates of recognition.



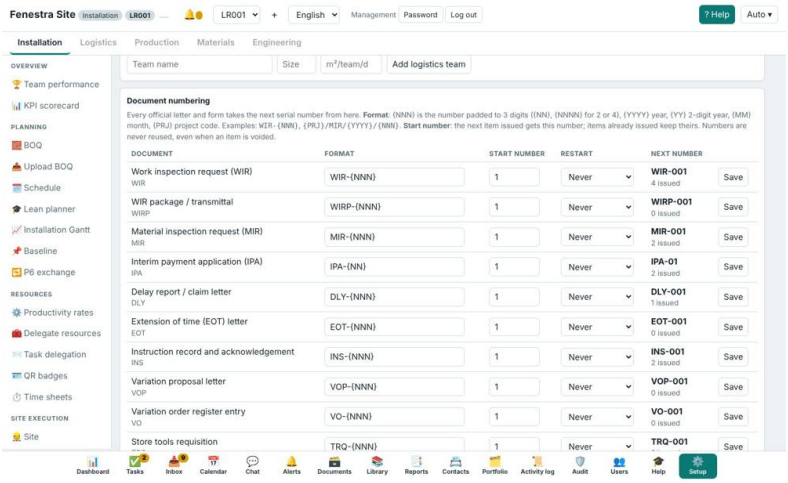
Documents: the register with numbers, types, dates, subjects and status. Filter by type, status, date or search text.

Using the register

1	Open **Documents** in the footer. Use the type chips, the search box, the status and the dates to filter.
2	Open an item to download its copy (the Word, Excel or PDF original, or a PDF), or to void it. A voided item keeps its number and the number is never reused.
3	Download the list as a report: Excel, PDF or CSV, all copies as a ZIP, or all copies as one PDF.
4	Press New letter / form to issue a document of any type by hand. Delay letters, EOT letters, instructions, IPA and variations are visible only to Management and the project manager.

Numbering (Setup > Document numbering)

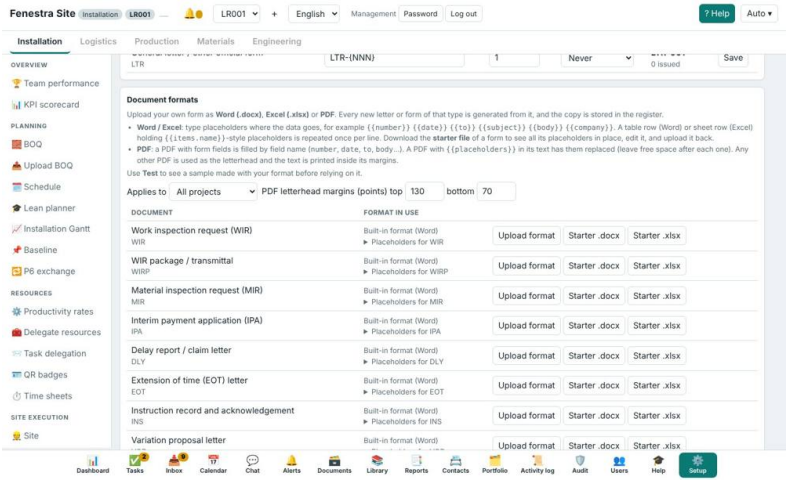
Each type has its own pattern. Use the tokens {NNN} (serial, with as many N as digits), {YYYY}, {YY}, {MM}, {DD}, {PRJ} (project code) and {TYPE}. Choose the reset (none, yearly or monthly) and the start number. A pattern needs a serial token. To start a type at a specific number, set the start before the first item is issued; to move an existing counter, enter the next number explicitly.



Setup: document numbering and the custom start number.

Formats (Setup > Document formats)

Upload your own Word (.docx), Excel (.xlsx) or PDF form for any type and the app fills it. In Word and Excel write fields as {{field}}; in a table row write {{list.column}} to repeat the row for each item. PDF forms are filled by field name. Download the starter file to see every field, and use Test to preview. Without an uploaded format the standard letter is used.



Setup: upload a Word, Excel or PDF format per document type.

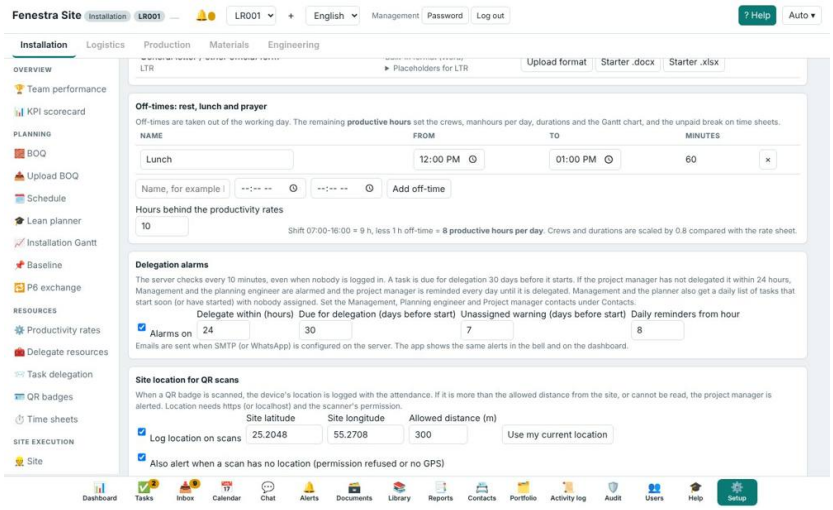
Word-to-PDF conversion is exact when LibreOffice is installed on the server; otherwise the PDF is a simplified rendering and the Word original is kept as the master copy.

12. Off-times, manhour forecast and operational budget

Who: Management sets the budget; project manager, planner and everyone see the forecast. Where: Setup, Dashboard.

Off-times

Rest, lunch and prayer breaks are not productive. In **Setup > Off-times** list each break with its start and end. The remaining **productive hours per day** drive the crews and manhours per day, the durations in the schedule and the Gantt chart, and the time sheets deduct the same breaks. 'Hours behind the productivity rates' (default 10) is the reference day on which the standard rates were measured, so a shorter productive day lengthens durations in proportion.

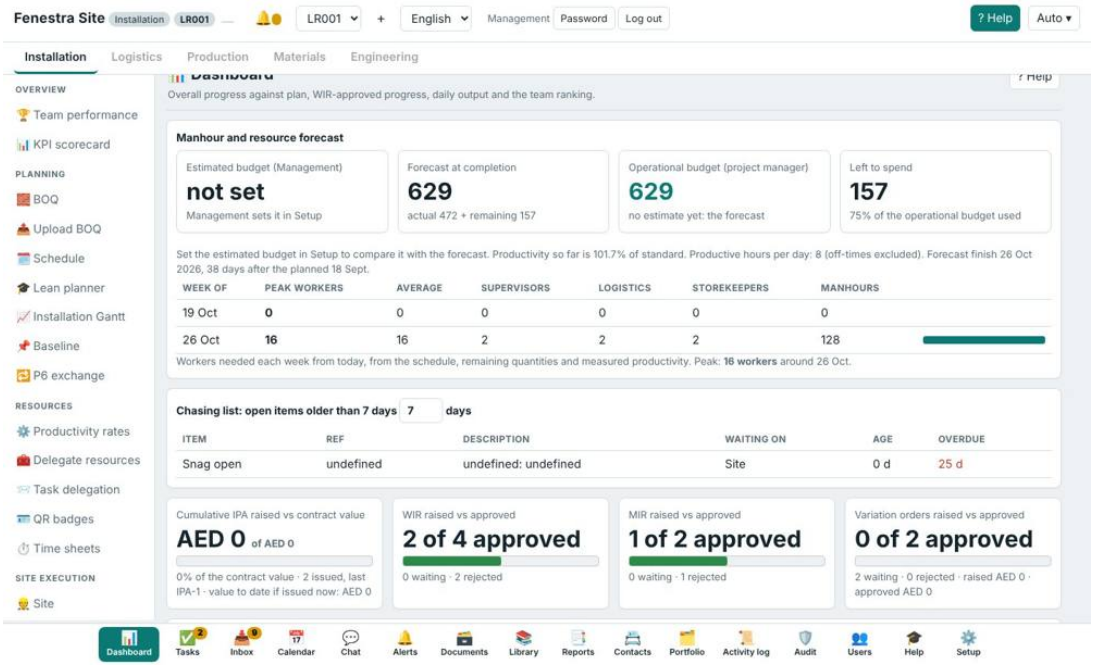


Setup: off-times and the productive hours per day.

Forecast and operational budget

The Dashboard card **Manhour and resource forecast** projects the manhours at completion (actual so far plus what remains at the measured productivity) and the workers needed each week by trade: workers, supervisors, logistics and storekeepers.

- **Estimated budget**: set by Management only (Setup).
- **Forecast at completion**: calculated automatically.
- **Operational budget**: the lower of the two. This is the budget the project manager works to, and the one used for the 50% and every-10% consumption alarms.
- If the forecast is higher than the estimated budget, Management and the Estimation manager are alerted by notification and email. Set both contacts in Setup > Contacts.



Dashboard: estimated budget, forecast, operational budget and weekly workers needed.

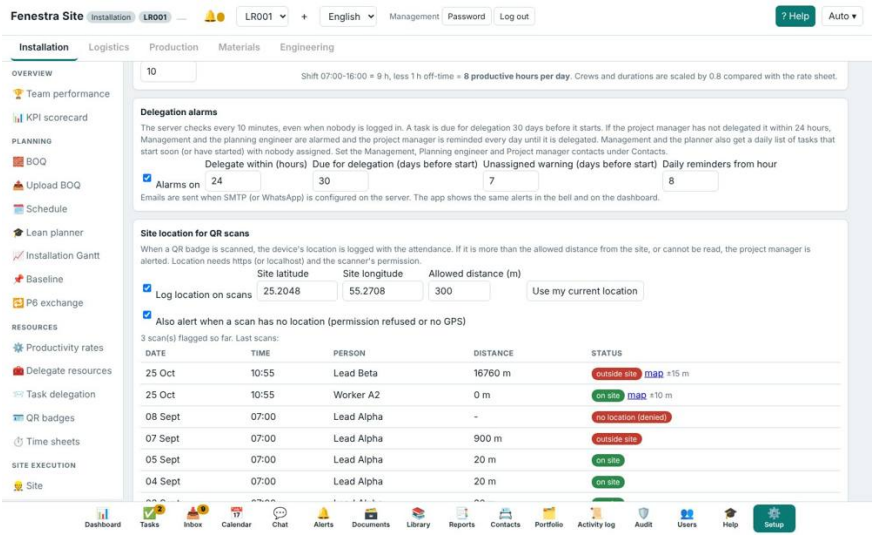
13. Delegation alarms and unassigned tasks

Who: Management, planning engineer and project manager. Where: Setup > Delegation alarms, Delegate resources, Tasks.

A task is due for delegation a set number of days before it starts (default 30, the same as the Delegate resources tab). From that moment the project manager has 24 hours.

Situation	What happens
Task not delegated within 24 hours	Management and the planning engineer get an alarm (once per task).
Task still not delegated	The project manager is reminded every day, from the reminder hour, until the task is delegated.
Task starts within 7 days (or has started) with nobody assigned	Management and the planning engineer get a daily list of unassigned tasks until every task has a site engineer.

The alarms are checked by the server every 10 minutes, so they go out even when nobody is logged in. Email and WhatsApp are used as configured; the same alerts appear in the Inbox and in Tasks. In Setup you can switch the alarms on or off and change the 24 hours, the 30 days, the 7 days and the reminder hour. Make sure Management, the planner and the project manager have an email address under Users or Contacts.



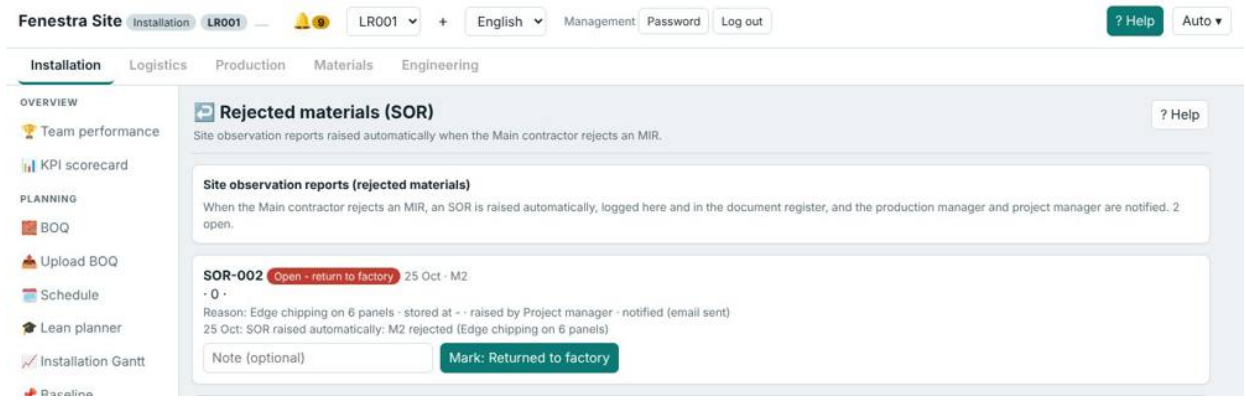
Setup: delegation alarms and the site location for QR scans.

14. Rejected material: site observation reports (SOR)

Who: project manager, production department, quality engineer. Where: Rejected materials (SOR), Documents.

When the Main contractor rejects a MIR, the app raises a **site observation report** automatically for the material to be returned to the factory for rectification. The report records the material, quantity, supplier, delivery note, MIR number, reason for rejection, the action required and where the material is stored.

1	The MIR decision is logged as Rejected in Materials.
2	An SOR is created, numbered (for example SOR-001) and logged in the document register with its copy.
3	The production manager and the project manager are notified (notification and email).
4	The production department updates the status on the Rejected materials tab: *Open - return to factory*, then *Returned to factory*, *Rectified at factory*, *Re-delivered to site* and *Closed*, adding a note each time (the button shows the next status).
5	The re-delivered material goes through the normal delivery, QA/QC and MIR steps again.



Rejected materials: each SOR with its reason, location and status buttons.

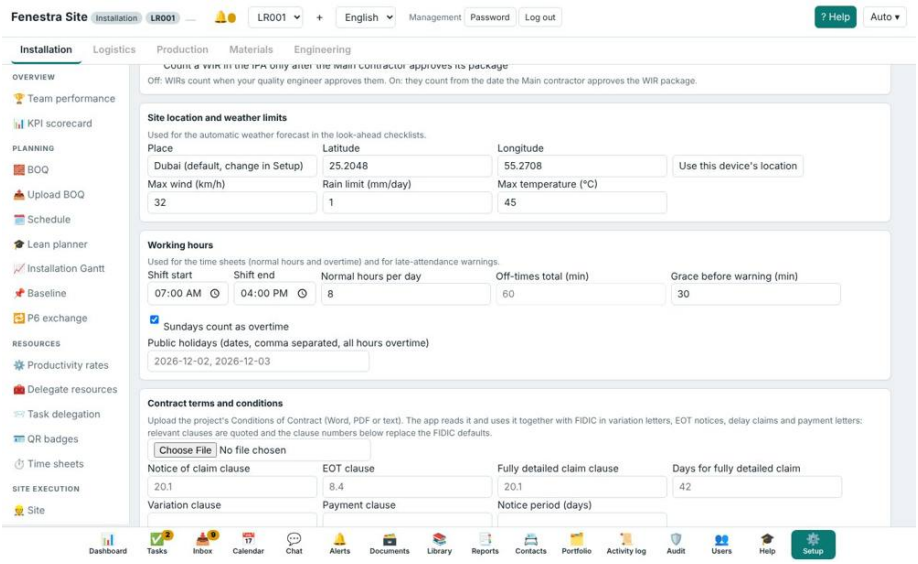
Open SORs also appear in Tasks for the project manager and the production department, and count in the QA/QC engineer KPIs.

15. Location of QR scans

Who: team leaders scan; the project manager is alerted. Where: Setup > Site location for QR scans.

Every QR scan by a team leader logs the device location with the attendance. The project manager is alerted when the scan is farther from the site than the allowed distance, or when the location cannot be read (for example the permission was refused).

1	In Setup > Site location for QR scans, enter the site latitude and longitude, or press **Use my current location** while standing at the site.
2	Set the allowed distance in metres (default 300) and tick whether to log locations and whether to alert when no location is available.
3	Team leaders must allow location access in the browser when they first scan. Location works only when the app is opened over https (or on localhost).
4	Flagged scans are listed under the same card with the person, the distance and a map link; each also raises a notification to the project manager.



Setup: site location, allowed distance and the list of flagged scans.

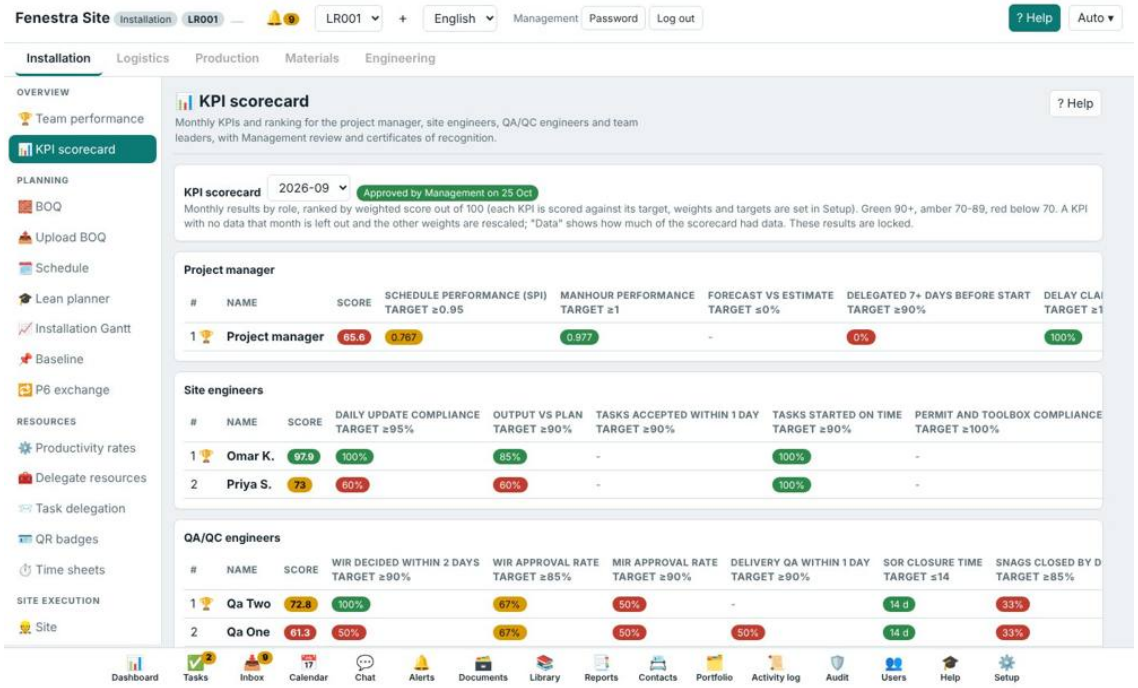
The location is only as accurate as the phone gives. Indoors it can be off by tens of metres, so keep the allowed distance generous.

16. Monthly KPIs, ranking and certificates of recognition

Who: Management reviews and approves; other roles see the approved results of the groups below them. Where: KPI scorecard (left menu, Overview), Setup > KPI targets and weights.

Every month the app scores four site groups from the data it already holds: the **project manager**, the **site engineers**, the **QA/QC engineers** and the **team leaders**, and four engineering groups (chapter 19). People see only the groups below their own role, never their own group or a superior's, and the Main contractor sees none. Each KPI is compared with a target, scored from 0 to 100 and weighted into one score, shown green, amber or red. People are ranked inside their group, with a six-month trend.

Group	KPIs (examples of what is measured)
Project manager (project level)	Schedule performance (SPI), manhour performance, forecast against the estimate, tasks delegated 7 or more days before start, delay claim letter on time, IPA issued on time, instruction proposals on time.
Site engineers (by tasks assigned)	Daily update compliance, output against plan, tasks accepted within 1 day, tasks started on time, permit and toolbox compliance.
QA/QC engineers (by name on inspections)	WIR decided within 2 days, WIR approval rate, MIR approval rate, delivery QA within 1 day, SOR closure time, snags closed by the due date.
Team leaders	Productivity index, output against plan, scan-in compliance, scanned on site, right first time (WIR) and safety and discipline (incidents plus warning letters).



KPI scorecard: ranking, score chips, each KPI against its target and a comment box per person.

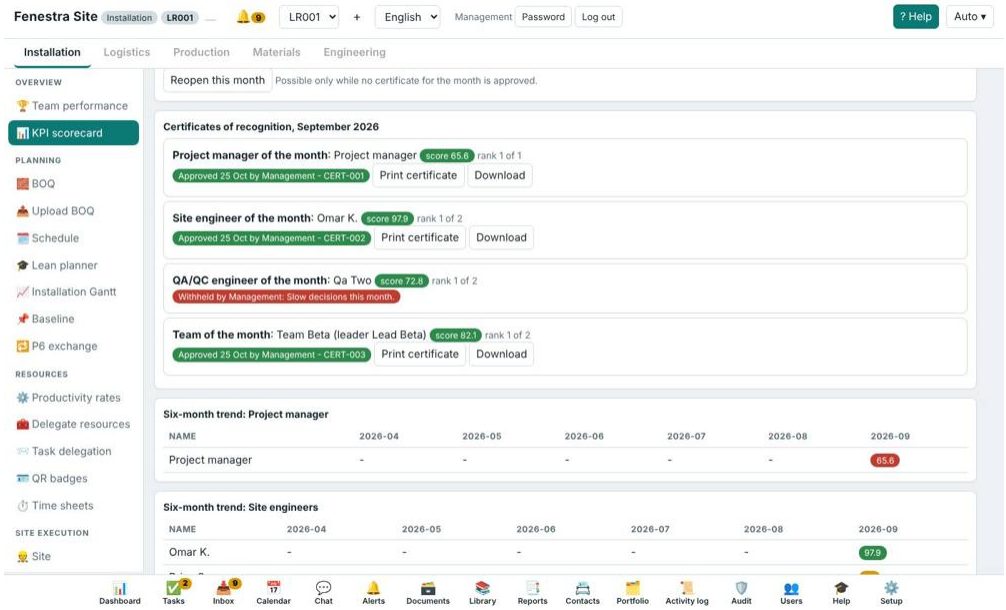
The monthly cycle

1	At the start of the month Management is notified that the previous month is ready. Everyone else sees it only as Draft (project manager) or not at all until it is approved.
2	Management opens **KPI scorecard**, selects the month and reads the ranking. A KPI with no data that month is left out and the other weights are rescaled.
3	Management types a comment for any person and an overall review note, then presses **Save review**. This stores a snapshot. Approval stays disabled until a review is saved.
4	Management presses **Approve results officially**. The results are locked: later data or target changes do not alter them, and the month is published to the others (the project manager also sees the comments). **Reopen** is possible until a certificate has been approved for the month.

Certificates of recognition

For each group the top-ranked person whose score reaches the certificate threshold (Setup) is proposed. Only Management can decide.

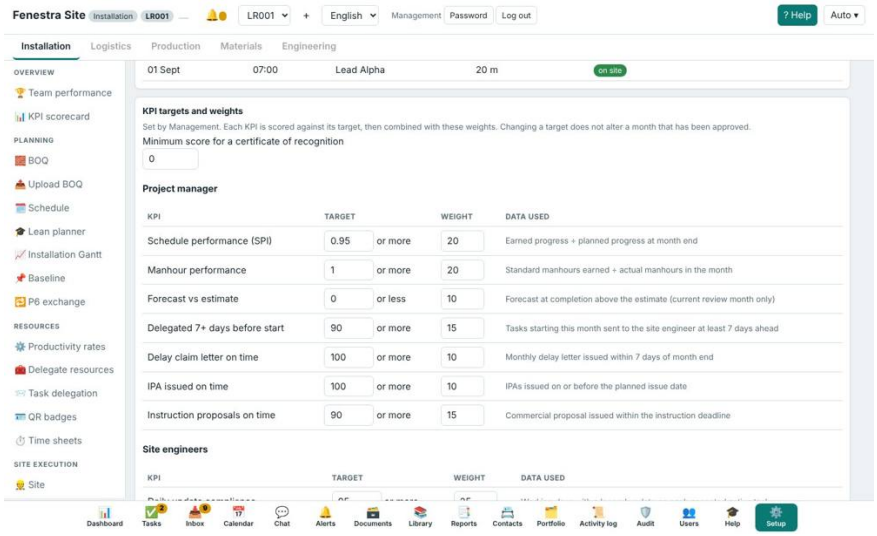
1. Under **Certificates of recognition** add an optional note and press **Approve certificate**. The certificate receives its number (CERT-001, CERT-002 and so on), is stored in the document register with a copy and a notification goes to everyone.
2. To refuse, write the reason in the note box and press **Withhold**. The reason is kept with the record and no number is used.
3. Press **Print certificate** and use Print or Save as PDF in the browser, or **Download** to keep the file. The certificate carries the company name and logo, the person or team, the award, the month, the score, the rank, the number and Approved by Management.



Certificates of recognition for the month, approved and withheld.

Targets and weights

Management sets each target, weight and the certificate threshold in **Setup > KPI targets and weights**. Other roles can read them but not change them. Changing a target never changes a month that is already approved.



Setup: targets and weights for every KPI.

The project manager scorecard is project level, so it ranks 1 of 1 and relies on the trend. QA/QC ranking uses the names recorded on inspections. If only role labels were recorded, it becomes a single entry.

17. Engineering module: roles and the working order

Who: Engineering manager, engineering team leaders, designers, document controller, Project manager, Procurement manager. Where: the Engineering module tab in the header.

The Engineering module takes the project from the handover of the contract documents to released shop drawings, take-offs, cutting lists and as-built drawings. Everything the engineering department produces is checked twice before it leaves engineering: first by the **engineering team leader** and then by the **Engineering manager**. There is no separate quality engineer in engineering. Wherever this chapter says **automatic**, the system does it without anyone pressing a button.

Roles

Role	What the person does
Project manager	Completes and shares the handover; reviews the contradictions; approves, returns or compresses the engineering programme; delegates the site-dimension survey and the handover documents.
Engineering manager	Sets the team structure in Setup; delegates the study and every submittal to team leaders; quality checks everything before release; acts on each contradiction (RFI, resolved, not an issue); approves output; closes RFIs.
Engineering team leader	Accepts tasks; delegates to designers by list or QR badge; checks quality and logs submittals, take-offs, cutting lists, GTOs and as-built drawings; re-delegates returned work; quality checks the designers' output.
Designer, Junior designer	Prepares the work and logs sqm issued with the man-hours taken.
Document controller	Submits quality-checked documents to the consultant, records the response and the RFI replies.
Procurement manager	Logs in to the Procurement tab: receives the BMTO, the MTO and the GTO, downloads each as CSV and ticks it when it has been sent to the vendor.
Planning engineer	Provides the baseline if missing; copied on programme and lapse notices.
Site engineer	Accepts the site-dimension survey or rejects it with a justification and a photograph.
Quality engineer	Prepares the installation-side handover documents delegated by the Project manager. Not part of engineering quality checks.
Management	Resource requests, KPI review, and the switch for the optional AI check in Setup.
Main contractor	Is told automatically when a site is not ready or when the IFC dimensions are to be followed. No KPI access.

Every delegation has a start and an end date

In both modules, no task can be delegated without a start date and an end date. The Engineering screens prefill the dates: the start is today and the end follows the duration set for that task type in **Setup > Engineering**. Change them if needed. Lapsed dates alert the Engineering manager, the Project manager, the planner, the team leader and the designers.

The working order

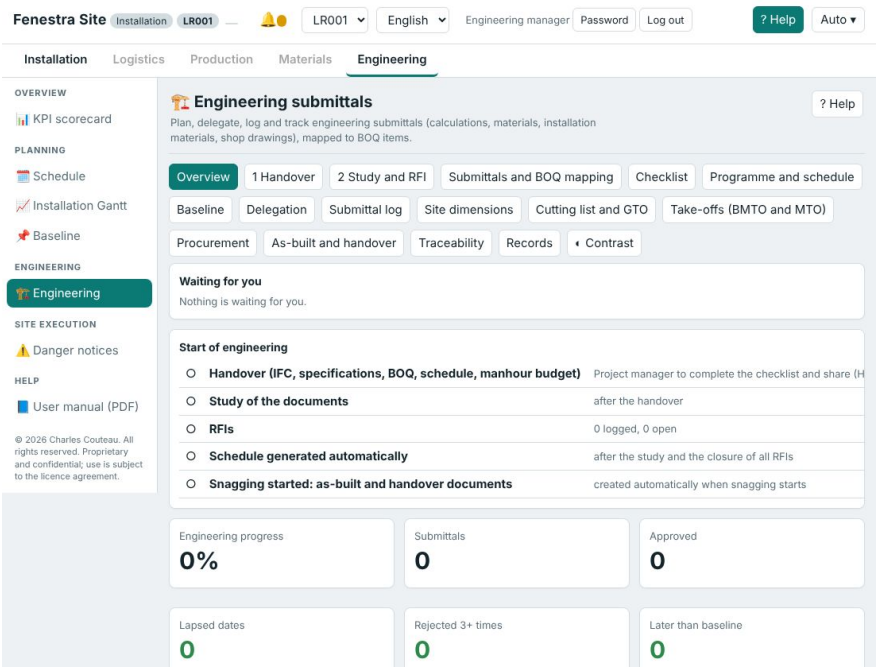
1	**Handover.** The Project manager completes the handover checklist (IFC drawings, specifications, BOQ, baseline schedule, engineering budget in manhours) and shares it. Automatic: the Engineering manager, planner and document controller are notified, and the AI check starts if it is switched on.
2	**Study.** The Engineering manager delegates the study of the IFC drawings, specifications, BOQ and schedule to team leaders with dates, or does it himself. A team leader accepts, or rejects with a justification.
3	**Contradictions.** Contradictions between the documents are listed (by the optional AI check, by the automatic rules, or by hand). The Project manager confirms or dismisses each item; the Engineering manager raises an RFI, resolves it or dismisses it. The study cannot be completed while an item is open.
4	**RFIs.** RFIs are logged with a reply deadline from the contract (14 days if none). The document controller records the reply; the Engineering manager closes it. Lapsed deadlines alert the Engineering manager, Project manager and planner.
5	**Schedule.** Automatic: when the last RFI closes the schedule is generated: material, structural calculation and concept drawing in parallel, then the shop drawing. The Engineering manager compares it with the baseline and gives a written justification if it is later.
6	**Programme decision.** The Project manager approves, returns it, asks Management for resources or orders a compressed revision.
7	**Delegation.** The Engineering manager delegates each submittal to a team leader with dates. The team leader accepts and delegates to designers by list or QR badge, with dates.
8	**Logging and quality.** The team leader checks quality, ticks the box and logs the document. The Engineering manager quality checks it. Only then can the document controller submit it to the consultant and record the response.
9	**Rejection or return.** A rejected or returned document goes back to the team leader who logged it. He re-delegates the rework (list or QR badge, with dates), possibly to another designer, and resubmits. More than two rejections raise an alarm.
10	**BMTO.** When a material submittal is approved the BMTO (bulk material take-off) task is created automatically. The team leader prepares it with the checklist (factory materials, site materials, consumables); after the two checks it goes to the Procurement manager to negotiate with vendors and conclude the supply agreement.
11	**Shop drawing.** It can be delegated only after the material, calculation and concept drawing are approved. It follows steps 7 to 9.
12	**MTO and site dimensions.** Automatic: when the shop drawing is approved, the MTO task and the site-dimension tasks are created. The MTO is taken off the released BMTO: every line names a BMTO and cannot exceed its quantity; it has a delivery date. The Project manager delegates the survey to a site engineer with dates, or instructs the Engineering manager to follow the IFC dimensions.
13	**Cutting lists and GTO.** When dimensions are received, cutting lists (split by product family) and the GTO (glass take-off) are created. They are delegated with dates, prepared with man-hours, checked by the team leader and then by the Engineering manager. The GTO then goes to the Procurement manager. Cutting-list materials lists are not sent to procurement.
14	**As-built and handover.** Automatic: when snagging starts, as-built tasks are created per BOQ item and the seven QA/QC handover documents for the Project manager to delegate.
15	**Output and productivity.** Designers log sqm issued with man-hours. The team leader checks it, the Engineering manager approves it, and it is ranked by sqm per man-hour.
16	**KPIs.** Automatic: monthly KPIs, ranking, Management review and certificates (chapter 19).

Every engineering notification must be acknowledged by its recipient and is kept as a permanent, hash-chained record (Records tab). Alarms are acknowledged one by one.

18. Engineering: screens and daily use

Who: everyone in the Engineering module. Where: Engineering tab, then the tabs along the top.

The Engineering overview opens with **Waiting for you**: tiles showing what is waiting for your role (to accept, to delegate, returned for rework, to quality check, to approve, to reply, lapsed). Press a tile to open the right tab. The tabs are grouped by stage: **Overview**, **Start** (handover, study and RFI, submittals, checklist), **Plan** (programme, baseline, delegation), **Deliver** (log, site dimensions, cutting list, take-offs, procurement) and **Close** (as-built, traceability, records). Choose a stage and its tabs appear in a second row. Your notifications to acknowledge are collapsed into one line, **N notifications to acknowledge**; press it, or the red tile, to open the list. Below are the start-of-engineering checklist, the pipeline, the items waiting, the budget chart and the ranking.



Engineering overview: Waiting for you, start of engineering and notifications.

Tab	What it is for
Overview	Waiting for you, pipeline per BOQ item, who is holding what, man-hours against the budget, flagged output, ranking, weekly report.
1 Handover, 2 Study and RFI	The handover checklist, the study tasks, the contradictions (cards with the two documents side by side) and the RFI log.
Submittals and BOQ mapping, Checklist	The planned submittals (material, structural calculation, concept drawing, installation material, shop drawing) mapped to BOQ items, and the checklist of open gaps.
Programme and schedule, Baseline	The generated schedule, the check against the baseline, the justification and the Project manager's decision.
Delegation	Engineering manager to team leader; team leader to designers (list or QR badge); acceptance; returned work.
Submittal log	The team leader logs documents; the Engineering manager quality checks; the document controller records the consultant response; output logging; history.
Site dimensions	Survey tasks, acceptance or rejection with photographs, or the IFC instruction.
Cutting list and GTO, Take-offs (BMTO and MTO)	Preparation, delegation with dates and QR, man-hours, the two quality checks and release.
Procurement	Released BMTO, MTO and GTO with lines and delivery date, CSV download, sent-to-vendor tick.
As-built and handover	As-built tasks and the seven handover documents for QA/QC.
Traceability, Records	BOQ item to every document with CSV download; the permanent engineering records with chain verification.

Delegating by list or by QR badge

A team leader chooses a designer from his team or scans the designer's QR badge, and gives a start and an end date. For work returned for rework, the card **Returned for rework** lets him pick one designer and re-delegate every returned item in one step, or scan one badge for all of them.

Quality checks

The team leader ticks that he has checked the document before logging it. The Engineering manager then works through the **Quality check queue** in the overview: tick the clean items and press **Pass all ticked**. To return an item, write the reason and press **Return**; it goes back to the team leader who logged it. Nothing reaches the consultant, the Procurement manager or the ranking before the Engineering manager's check.

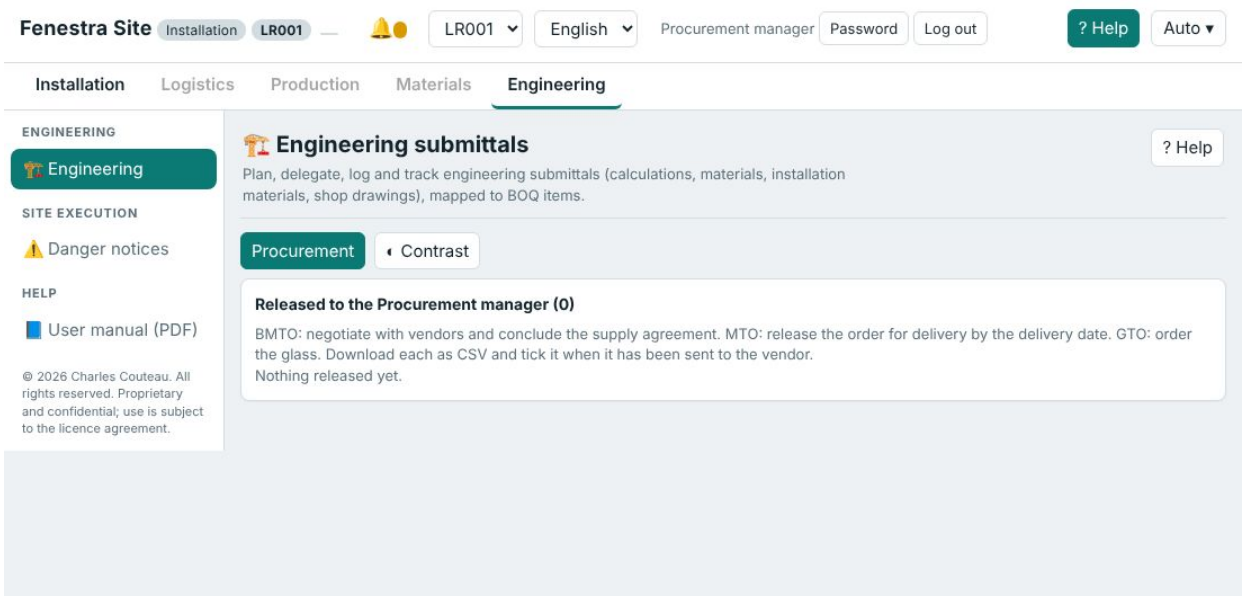
Logging output with man-hours

Designers log the sqm issued and the man-hours taken; man-hours are required. **Same as last** copies your previous hours and **Start timer** / **Stop timer** enters the hours for you. Output that looks odd (very high sqm per man-hour, or too many hours for one person in a day) is accepted but flagged for the checkers. The team leader checks the output, the Engineering manager approves it, and only then does it count in the ranking.

BMTO, MTO and GTO

Document	Created	Goes to the Procurement manager for
BMTO	When a material submittal is approved	Negotiating with vendors and concluding the supply agreement.
MTO	When the shop drawing is approved; taken off the released BMTO	Releasing the order for delivery by the delivery date.
GTO	When dimensions are received (replaces the GPO)	Ordering the glass.

In the **Procurement** tab the Procurement manager sees what has been released, with the lines and the delivery date, downloads each document as a CSV file and enters the vendor or order reference when it has been sent. Engineering is notified.



Procurement tab: released BMTO, MTO and GTO with the sent-to-vendor tick.

Pipeline, waiting list, budget and weekly report

The **pipeline** shows one row per BOQ item with a mark for each stage: green done, amber in progress or waiting for a check, red rework or lapsed, grey not started. **Who is holding what** shows the person, the step and the days waiting; items waiting more than the days set in Setup (default 2) turn red. The budget chart compares the man-hours logged with the budget and the plan. **Weekly report** opens a one-page report to print or save as PDF.



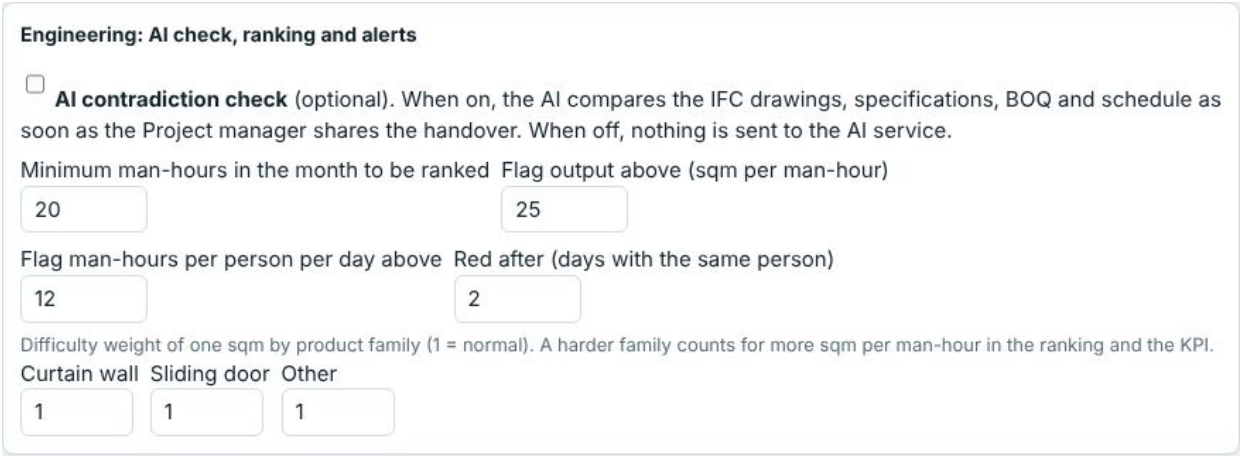
Pipeline per BOQ item and the items waiting.

The optional AI contradiction check

The AI check is **off by default**. The Engineering manager or Management switches it on in **Setup > Engineering: AI check, ranking and alerts**. When it is on, the AI compares the IFC drawings, specifications, BOQ and schedule as soon as the Project manager shares the handover, and lists the contradictions for review. Select the IFC and specification files (PDF, text or image) from the Library; the screen shows which files and how many pages the AI read, and warns if it read none. When it is off nothing is sent to the AI, the BOQ-and-schedule rules check can be run by hand, and contradictions can be listed manually. The administrator must set **ANTHROPIC_API_KEY** on the server (Appendix F).

On a phone and in bright light

On a phone the controls are larger, the tab bar scrolls sideways and tables are shown as cards. Long cards fold; press the heading to open them. **Contrast** in the tab bar switches a high-contrast display on and off on that device.



Setup: AI check switch, ranking rules and alert settings.

19. Engineering KPIs, ranking and who sees them

Who: Management reviews and approves; each person sees the groups below his own role. Where: KPI scorecard (Overview) and Setup > KPI targets and weights.

The scorecard of chapter 16 also covers four engineering groups. They use the same monthly results, ranking, Management review, approval lock, certificates and Setup targets and weights.

Group	KPIs measured
Engineering manager (department)	Submittals on time, baseline compliance, first-time approval, approval cycle time, lapsed items, delegation speed, cutting lists, GTO and as-builts on time, RFIs answered by the deadline, man-hours against the handover budget.
Engineering team leaders	Delegated work on time, first-time approval, rework turnaround, acceptance response, team productivity in sqm per man-hour, acknowledgement of notifications.
Designers	Quality-adjusted sqm per man-hour (the main KPI), delivered inside the window, output passing the team leader's and the Engineering manager's checks, acknowledgement of notifications.
Junior designers	The same KPIs as designers; there are no lower sqm targets.

Productivity: sqm per man-hour

Productivity is approved sqm divided by the man-hours logged. With no target set, it is scored against the best person in the group; set a target in Setup to score against a fixed number. In **Setup > Engineering** the Engineering manager can set the minimum man-hours in the month needed to be ranked (default 20) and a difficulty weight per product family (curtain wall, sliding door, other; default 1). The same weights apply to the KPI. There are no fixed sqm targets.

Who sees which KPIs

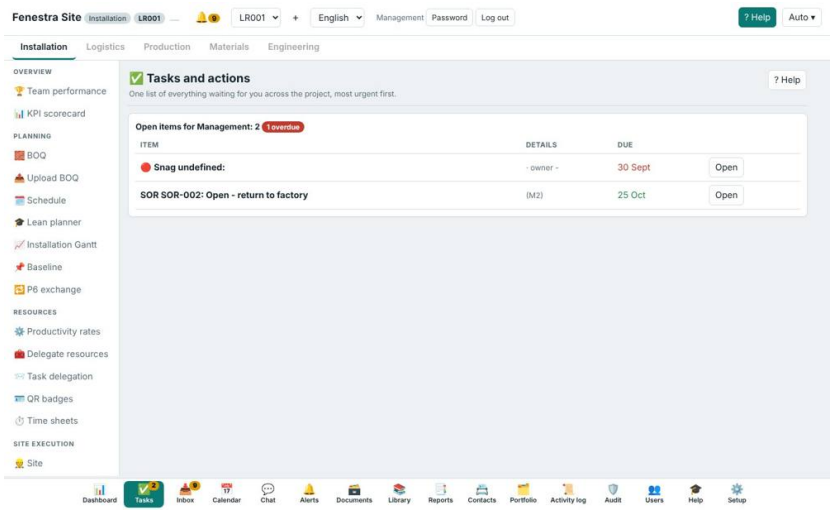
Role	KPI groups visible
Management	All groups.
Project manager	Site engineers, QA/QC, team leaders, Engineering manager, engineering team leaders, designers, junior designers (not his own group).
Planning engineer	Site engineers, QA/QC and site team leaders.
Engineering manager	Engineering team leaders, designers and junior designers.
Engineering team leader	Designers and junior designers.
Site engineer	Site team leaders.
Everyone else, including the Main contractor	None.

A superior sees the KPIs of the people below, never the reverse, and nobody sees his own group or a superior's. The **Look-ahead** KPI sits in the Planning group.

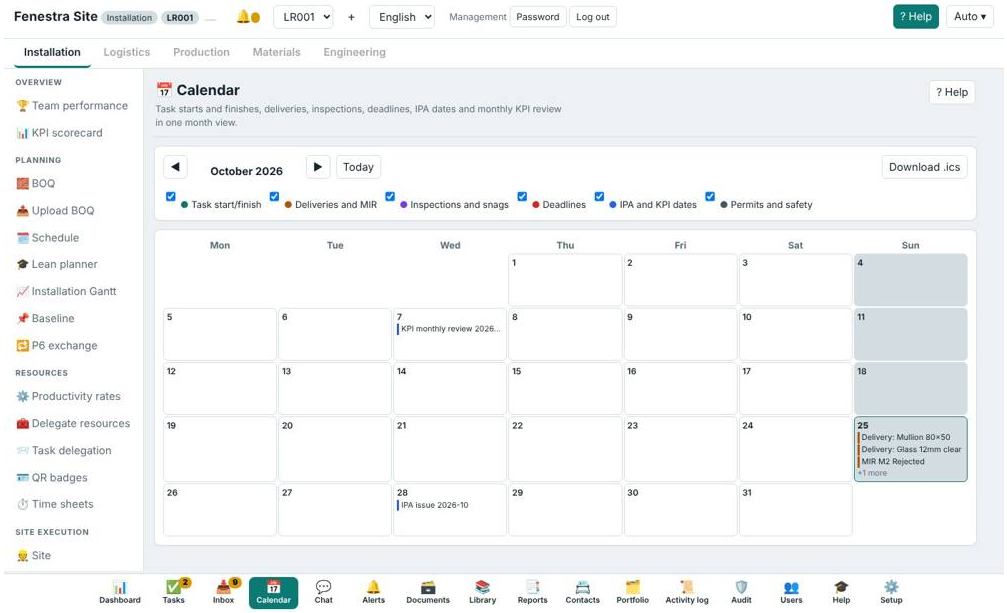
Appendix A. The common tabs (footer)

These tabs are not specific to the Installation module. Each person sees only those their role allows (appendix C).

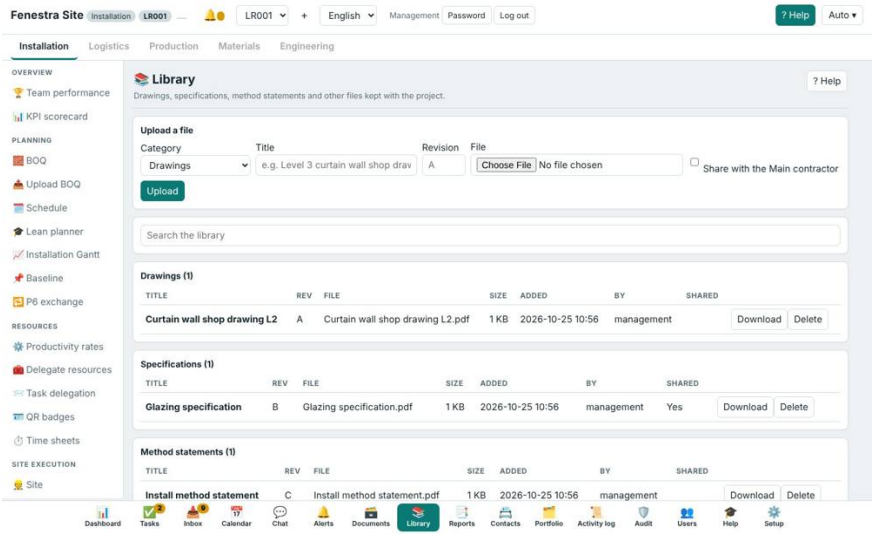
Tab	What it does
Dashboard	Plan versus actual, WIR-approved progress, daily output, manhour and resource forecast, chasing list, S-curve and charts.
Tasks	One list of everything waiting for you, most urgent first, with a button to open each item: delegations, unassigned tasks, WIR and MIR decisions, SORs, snags, requests, instructions, IPA drafts, permits, and for Management the KPI review and certificate decisions.
Inbox	Every notification sent to you with unread marks and filters (Alerts, Documents, KPI, Tasks, Other). Mark all as read or clear the read ones. The bell icon opens it.
Calendar	Month view of task starts and finishes, deliveries, MIR and WIR, inspections, deadlines, IPA cut-off and issue days, the KPI review, permits and toolbox talks. Switch types on or off and download an .ics file for your phone or Outlook.
Chat	Project chat for all parties, logged per project.
Alerts	Pre-start alerts and other flags, with a count badge.
Documents	The numbered register of letters and forms (chapter 11).
Library	Drawings, specifications, method statements, templates, reports and contracts, with title and revision, up to 8 MB per file. Management and the project manager can tick Share with the Main contractor, who then sees only shared files. You can delete your own files; Management and the project manager can delete any.
Reports	Daily diary, weekly and monthly reports, with print and Excel.
Contacts	The project contacts and the team with call, email and WhatsApp links, search and a CSV download.
Portfolio	Projects, sub-projects, buildings, scopes and zones as a tree with rolled-up totals. Management can delete, and Management or the project manager can archive, a project (see Appendix F).
Activity log	Every delegation, acceptance, revert, email and approval with date and time.
Audit (Management)	The server audit log (sign-ins, user changes, backups, numbering changes, uploads) and the sensitive project changes (rates, budgets, baselines, approvals, certificates). Search and download as CSV.
Users (Management)	Create users, reset passwords, change roles, plus a table showing which tabs each role can open.
Help	Searchable guides, a quick-start checklist for your role, common questions and print.
Setup	Company details, users and sign-ups, permissions, contacts, budget, IPA, off-times, numbering, formats, delegation alarms, site location, KPI targets, email and WhatsApp, backups.



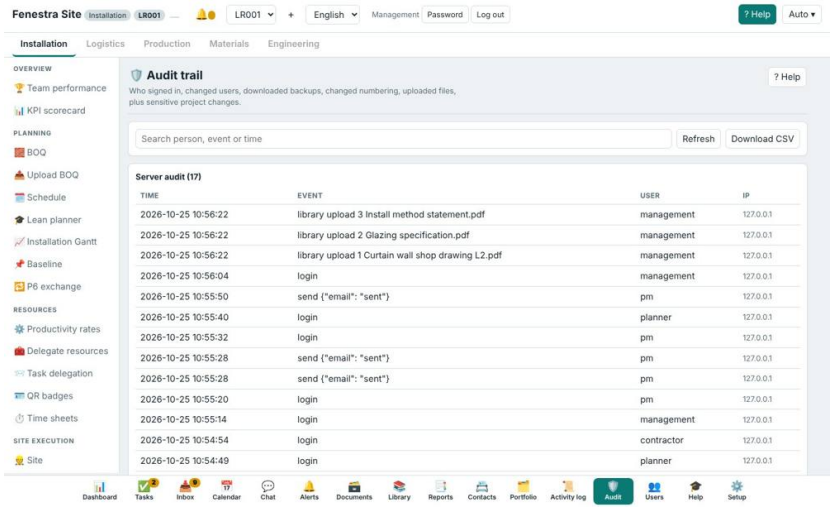
Tasks: open items for your role, overdue first.



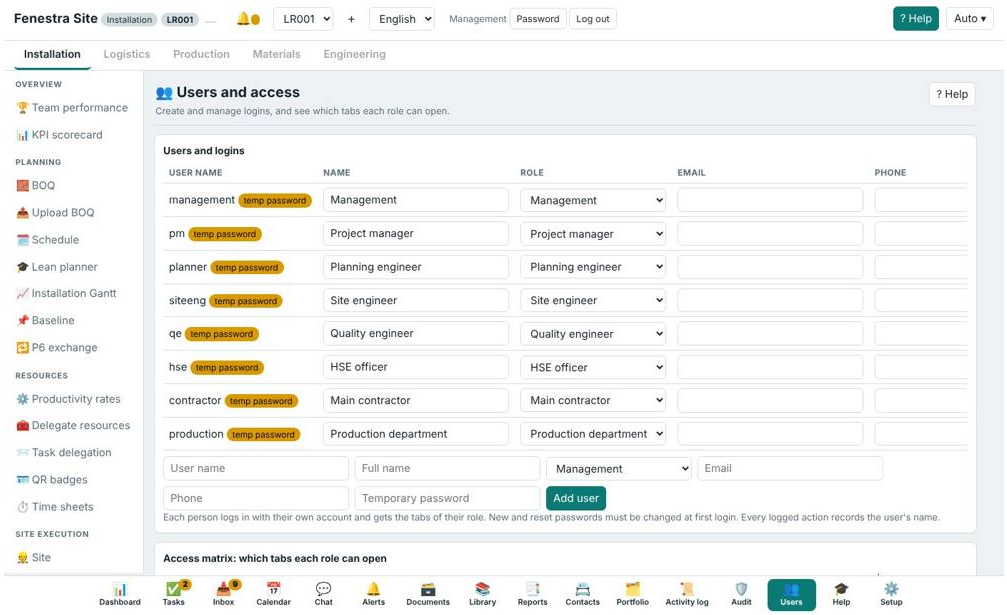
Calendar: one month with filters and .ics download.



Library: upload, search and share project files.



Audit trail (Management only).



Users and access (Management only).

Appendix B. Other tabs

Tab	What it does
Lean planner	Optional guided Last Planner System: value and waste, takt zones, trade train, pull planning, make-ready and weekly PPC. The project manager or planning engineer chooses Yes or No. When on, the schedule is locked and changed only through the planner, with a reason for each change.
Installation Gantt / Baseline	Bar chart of the programme, and the frozen baseline for comparison.
Productivity rates	Standard quantity per team per day and team sizes, with an approval workflow.
Site tools	List the tools required, plan their disbursement to installers by employee ID, and get a store tools requisition form automatically. Mark tools issued and returned. The tool assignment log downloads as Excel.
Commercial	Contract rates and estimate cost, costs, labour cost per manhour, retention and the older progress invoice (visible to Management and the project manager).
Claims / Delay analysis	Extension of time and delay documentation (see chapter 10).
Team performance	Monthly team ranking, points for on-time tasks, congratulation pop-ups and the team-of-the-month recommendation.
KPI scorecard	Monthly KPIs, ranking, review and certificates (chapter 16).
Rejected materials (SOR)	Site observation reports for rejected material (chapter 14).
Progress photos	Upload photos (compressed automatically) tagged by building and elevation; slide through them per elevation. A reminder goes out every Saturday.
Safety / Snagging	Permits, toolbox talks, incidents; snag list with drawing pins.
Look-ahead	The look-ahead lists and confirmations.

Appendix C. Alarms and automatic actions

Event	What the app does automatically
Activity starts within 3, 2 or 1 month	Pre-start alert for each unmet prerequisite; missing items trigger an email to the Main contractor or Production.
Task due for delegation, not delegated in 24 hours	Alarm to Management and the planner; daily reminder to the project manager until delegated (chapter 13).
Task starting soon with nobody assigned	Daily list to Management and the planner until assigned.
Task delegated	Invitation to the site engineer; accept or revert is logged.
Handover shared (engineering)	The Engineering manager, planner and document controller are notified; the study tasks are created; the AI check starts if it is switched on.
Document logged, take-off or cutting list submitted	The Engineering manager is told that a quality check is required; the item is released only after the check.
Rejection or return	The team leader who logged it is told to re-delegate with dates; more than two rejections alarm Management and the Project manager.
Shop drawing approved	The MTO task and the site-dimension tasks are created; the Project manager and site engineers are notified.
BMTO, MTO or GTO released	The Procurement manager is notified; engineering is notified when it is sent to the vendor.
Engineering notification	Stays with the recipient until acknowledged; alarms are acknowledged one by one.
Task finished on or before the planned date	Congratulation pop-up and points for the team; monthly ranking and team-of-the-month report.
No daily update on a started activity	Alarm for yesterday, and for today from 17:00; list on the project manager's dashboard.
Task past its planned finish with quantity remaining	Recorded as carried over to the next day with the same team; forecast finish and delay shown on the Schedule and Gantt.
Manhours reach 50%, then every 10%	Notification and email to the project manager against the operational budget.
Forecast manhours above the estimated budget	Alert to Management and the Estimation manager.
Material delivery approved by QA/QC	MIR raised and a letter sent to the Main contractor.
MIR rejected by the Main contractor	SOR raised, numbered and logged; production manager and project manager notified (chapter 14).
QR scan away from the site, or without location	Project manager notified and the scan flagged (chapter 15).
Instruction recorded	Acknowledgement letter to the Main contractor, copy to Engineering and Commercial with a 7 working day deadline; BOQ line added under Variations; alert if the deadline is missed.
Variation schedule ends after baseline completion	Extension of time letter issued.
Tool disbursement planned	Store tools requisition form created and emailed to the store keeper.
IPA issue day	Draft monthly IPA prepared and the project manager is notified.
Start of each month	Management is told the previous month's KPIs are ready for review.
Certificate approved or withheld	Numbered, logged in the register and notified to everyone (chapter 16).
Every Saturday	Reminder to site engineers to upload progress photos.
Item waiting longer than the chase period	Notification and a line in the dashboard chasing list.
Cutting-list item short near its needed-by date	Notification and a Needed soon / overdue flag.
Tool not returned by its return date	Daily notification and an overdue flag.
Request overdue	Escalation text to the Main contractor's escalation contact.

Emails and WhatsApp messages are sent by the server only if your administrator has configured them. Otherwise the letter text is kept with the record so that it can be copied and sent manually.

Appendix D. Who sees what

Role	Main tabs
Management	Everything, including Audit, Users, KPI review and approval, certificates, the estimated manhour budget and Permissions in Setup.
Project manager	Everything except user administration and the audit trail; sets locations, checklists, IPA settings and the operational budget; sees rates, IPA, delay letters, EOT letters and instructions; sees KPI drafts and comments.
Planning engineer	Planning, schedule, Lean planner, P6, delay analysis and the common tabs; does not see rates, IPA, commercial values, delay letters, EOT letters or instructions.
Site engineer	Schedule, Gantt, Baseline, Site, WIR log, Snagging, Safety, Materials, Site tools, Progress photos, Look-ahead, plus the common tabs (Tasks, Inbox, Calendar, Chat, Alerts, Documents, Library, Contacts, Activity log, Help) and approved KPI results.
Quality engineer	Materials, WIR log, Snagging, Rejected materials and related quality tabs, plus the common tabs and approved KPI results.
Engineering manager, Engineering team leader, Designer, Junior designer, Document controller	The Engineering module (chapters 17 and 18), the dashboard, planning tabs and the common tabs; the Engineering manager also has Setup. KPI groups as in chapter 19.
Procurement manager	The Procurement tab of the Engineering module and the common tabs; receives the BMTO, MTO and GTO. No KPI tab.
HSE officer	Safety and site records, plus the common tabs.
Main contractor	Dashboard, Schedule, Installation Gantt and Baseline (read only), Inbox, Help, and the Library files marked as shared. No KPI tab.
Production department	Chat, material requests and Rejected materials (SOR), plus Inbox and Help.

Management can change any role's tabs in Setup under Permissions. A role with custom tabs needs the KPI scorecard ticked there to see it. Delay letters, EOT letters and instructions stay limited to Management and the project manager whatever the permissions say.

Appendix E. Reports, signatures and chasing

Printable reports. At the bottom of the WIR log, Materials (MIR) and Snagging tabs, choose an item and press Print / save as PDF. The report holds the checklist results, quantities, dates, photos, decisions and signatures. Register (Excel) downloads the whole list, and Whole snag list prints every snag.

Signatures. In the same box draw your signature with the mouse or your finger and press Save. It is then stamped automatically, with your name and date, on the WIR decisions, QA/QC approvals, MIR decisions and snag fix or close-out that you sign. Press Remove to delete it.

Chasing list. The dashboard (project manager, Management, quality engineer) lists open items older than the number of days set on the card (default 7), oldest first: WIRs and deliveries awaiting QA/QC, WIR packages, MIRs and variation proposals awaiting the Main contractor, open snags, and variations whose details are overdue. Each item also raises one notification per period.

Materials short. Give each cutting-list item a Needed by date. An item that is still not fully delivered within 7 days of that date is flagged on the Materials tab and in the chasing list, with a notification.

Tools. Give a tool a Return by date when you plan it. A tool still out after that date, or after the default number of days (14) when no date is given, is flagged as overdue. The Site tools tab also shows which tools each installer holds.

Variation register. The Instructions tab starts with a register of all variations: status, days open, details due, value (Management and project manager only), time impact, extension of time noticed and totals, with an Excel download.

Appendix F. Administration and troubleshooting

First start and passwords

The first Management account is user name **management**. Its first password is the INIT_ADMIN_PASSWORD value that was set when the server was installed (or a random one printed once in the server log). You must change it at first login. If it is lost, delete users.json in the data folder and restart the server, which creates the management account again.

Data and backups

All project data now lives in one SQLite database, **fenestra.db**, in the server's data folder (the /data volume in Docker). It replaces the earlier Excel workbook per project; old workbooks are imported once at the first start and can then be deleted. The register copies, uploaded formats, the Library files and the notices already sent are stored in the same database. Users, company details and the audit log stay in their own files in the data folder.

Management can download a backup from Setup. It holds a consistent snapshot of fenestra.db and **fenestra.sql**, a plain SQL text dump that can restore the data (sqlite3 new.db < fenestra.sql). A daily backup is also made automatically. Back the volume up regularly.

Deleting projects and uploaded data

Delete a project (Management only): open the Portfolio tree and press **Delete** beside a project that has no sub-projects, then type the project code to confirm. The project's BOQ, programme, site records, library files, register documents and engineering records are removed. Before that the server saves a copy in the **deleted** folder of the data folder (a zip with the state, the library files and the register files), so an administrator can recover it. The last remaining project cannot be deleted. The daily backups still hold the project until they are replaced.

Delete uploaded data (Management and project manager): **Setup > Backup, archive and delete uploaded data** removes the BOQ, the scope-level programme, the contract or tender document text, or the imported cutting lists. Before each deletion a backup copy of the whole project is saved in the **archives** folder of the server (if the server cannot save it, you are asked whether to continue). Each deletion is recorded in the activity log. Library files are deleted in the Library tab (by the person who added them, the project manager or Management), document formats in Documents, and progress photos in the photo viewer.

Deleting a BOQ that already has site records. If site logs, inspections, obstructions, snags, delay records, variations, costs, requests, payment applications, deliveries, engineering submittals or recorded progress refer to the BOQ, the screen lists them and asks you to type **DELETE ALL**. The BOQ and all of those records are then deleted; the activity log, productivity rates, teams, contacts, library files, register documents and the engineering handover and study data stay. The backup copy taken just before holds everything that was deleted.

Archive a project (Management and project manager): press **Archive** beside a project in the Portfolio tree, or **Download an archive of this project** in Setup. You receive a zip with the project state, the library files, the register documents and the engineering records, and a copy is also kept in the **archives** folder on the server. Archives are for safekeeping and recovery by an administrator; the app does not restore them by itself.

Server settings

Setting	Purpose
DATA_DIR, PORT	Where the data is kept and the port the server listens on.
INIT_ADMIN_PASSWORD, SEED_DEMO_USERS	First Management password; set SEED_DEMO_USERS=0 in production so demo accounts are not created.
ANTHROPIC_API_KEY, ANTHROPIC_MODEL (optional)	Needed only for the optional AI contradiction check (and the in-app guide). Without the key the AI check is not available and the BOQ-and-schedule rules run instead.
TRUST_PROXY=1	Use the real client address behind an https reverse proxy.
SMTP_HOST (and related) or the WhatsApp variables	Needed for email and WhatsApp, including the delegation alarms, which run in watch.py inside the server.
https	Needed for location on QR scans (chapter 15) and for browser notifications.

The container image includes the extra modules docs.py (register, numbering, formats), watch.py (delegation alarms) and lib.py (Library), and the Python packages openpyxl, pypdf and reportlab. See DEPLOY.md for installation.

Common questions

Question	Answer
Why is the Schedule locked?	The Lean planner is on. Open the Lean planner and finish steps 1 to 4, or switch the Lean planner off there.
A WIR or MIR is not in the IPA.	Only approved items count, up to the cut-off day, and only for BOQ lines that have a rate. Check the MIR decision date and the Setup percentages.
The quantity was rejected.	A cumulative quantity cannot exceed the BOQ quantity. Check the BOQ line or split it correctly.
No emails are sent.	Ask the administrator to configure email; contacts and users must have an email address (Setup > Contacts, Users).
A tab is missing.	Your role does not include it. Management can change this in Setup > Permissions.
I cannot log in.	Check the user name and password. After several wrong attempts the account is locked for a short time. Management can reset passwords.
The Logistics, Production and Materials tabs do not open.	They are greyed out on purpose: those modules are not built yet. The Installation and Engineering modules are available.
The AI contradiction check does not run.	It is off by default. Management or the Engineering manager switches it on in Setup, and the administrator must set ANTHROPIC_API_KEY. Without it the BOQ-and-schedule rules run instead.
A designer is "not ranked".	The designer has fewer man-hours in the month than the minimum set in Setup (default 20).
Engineering says I cannot delegate.	Every delegation needs a start date and an end date, and the work must be accepted by the team leader first.
The KPI month shows Draft or is empty.	Only Management and the project manager see a month before Management approves it.
A certificate cannot be approved.	The month must be reviewed and approved first, and only Management can decide. The score must reach the threshold in Setup.
Scan locations are not logged.	The app must be opened over https and the team leader must allow location. Check Setup > Site location for QR scans.
The Library or Audit list looks old.	The list refreshes when you open the tab and every 15 seconds; press Refresh on the Audit tab.